

Program 3: Clustering

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Abstract

This report presents a clustering analysis of 11,500 EKG signal samples from 23 subjects across five activities. Each 1-second sample, consisting of 178 values, was transformed using feature extraction and normalization techniques to enhance cluster separability. For this report, we will be using **K-means**, **Agglomerative**, and **Gaussian Mixture Model** algorithms were evaluated, with clustering performance assessed using internal metrics and the Normalized Mutual Information (NMI) score for leaderboard submission. The approach highlights the effectiveness of preprocessing and model selection in unsupervised learning for time-series biomedical data.

Index Terms: K-means, Agglomerative, Gaussian Mixture Model.

1 Introduction

In this project, I clustered 11,500 EKG signal samples collected from 23 subjects performing five different activities. I began by preprocessing the data, applying normalization and transforming the signals using Fast Fourier Transform (FFT) to extract frequency-domain features. I then applied both K-means, Agglomerative, and Gaussian Mixture Model clustering algorithms to group the samples into 115 clusters. To evaluate performance, I used silhouette scores and submitted my results to the CLP leaderboard, where clustering quality was assessed using the Normalized Mutual Information (NMI) score.

2 Libraries and Environment Setup

The project was developed in a Python 3.8+ environment using Jupyter Notebook. Key libraries included NumPy and Pandas for data manipulation, SciPy for signal processing through Fast Fourier Transform (FFT), and scikit-learn for implementing clustering algorithms—specifically K-Means, Agglomerative, and Gaussian Mixture Models (GMM). Evaluation was performed using silhouette scores, and visualizations were generated using Matplotlib. StandardScaler from scikit-learn was used to normalize the features.

3 Data Exploration and Analysis

The dataset contains 11,500 samples and each has 178 features (EKG values).

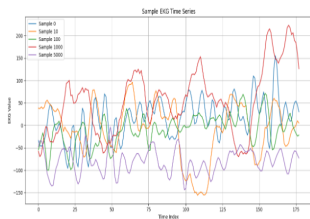


Figure 1. Sample Signals

Figure 1 displays five randomly selected EKG signal samples, each representing one second of cardiac activity captured as a 178-point time series. The x-axis corresponds to the time index, while the y-axis represents the EKG voltage values. Each line illustrates the unique waveform patterns of individual samples, highlighting dif-

ferences in amplitude and frequency characteristics. These variations indicate subject-specific or activity-specific signal dynamics, emphasizing the importance of feature extraction and normalization before applying clustering algorithms.

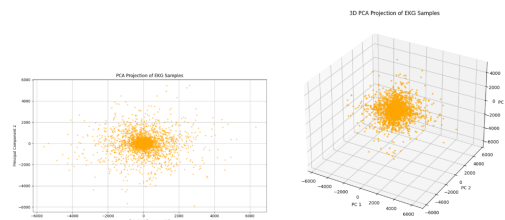


Figure 2. PCA Projection of EKG Samples

The two figures show the results of dimensionality reduction applied to the EKG dataset using Principal Component Analysis (PCA). The first figure is a 2D PCA projection, plotting each EKG sample along the first two principal components. The second figure extends this view into 3D by adding the third principal component. In both projections, we observe that the data forms a dense, spherical cluster centered near the origin, with no clear separation into distinct subgroups. This pattern suggests that the data distribution is continuous and overlaps heavily, rather than being made up of well-separated, compact clusters.

Given this observation, density-based clustering methods like DBSCAN may not perform well because they rely on the assumption of dense regions separated by low-density gaps, which this data lacks. K-Means may also struggle since it assumes spherical clusters of similar size and is sensitive to initialization and outliers. However, due to the compact and isotropic spread, Gaussian Mixture Models (GMM) become a suitable choice. GMM can model overlapping and elliptical distributions better by assigning probabilistic cluster memberships, making it more flexible for data like this where hard boundaries between clusters do not clearly exist.

4 Data Preprocessing

For the data preprocessing step, we applied **Fast Fourier Transform (FFT)** to each signal to convert it from the time domain to the frequency domain. This allowed us to extract frequency-based features, which are often more informative for

identifying physiological patterns in biomedical signals.

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-2\pi i \cdot kn/N}, \quad \text{for } k = 0, 1, \dots, N-1 \quad (1)$$

The Fast Fourier Transform (FFT) is an efficient algorithm for computing the Discrete Fourier Transform (DFT), which converts a time-domain signal into its frequency-domain representation.

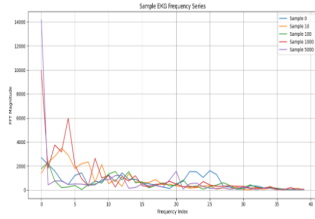


Figure 3. EKG Samples Converted into FFT

This plot shows the FFT magnitude spectrum of five EKG signal samples, capturing their frequency-domain representation. Each line corresponds to a sample's FFT magnitudes plotted against frequency index. The strong peaks near the lower frequencies (left side of the graph) indicate that EKG signals are dominated by low-frequency components, which is typical of biological signals like heart activity.

5 Clustering Algorithms

In this section, we evaluate the performance of three clustering algorithms—K-Means, Agglomerative, and Gaussian Mixture Models (GMM)—using both the similarity matrix and the Silhouette score as evaluation metrics.

5.1 K-means Clustering

In this section, the K-means clustering algorithm was applied to the PCA-reduced representation of the scaled version of the original dataset. The number of clusters is set to $k=115$. The model was initialized with a fixed random seed for reproducibility and run with 10 initializations to improve centroid stability.

```
k = 115
kmeans = KMeans(n_clusters=k, random_state =
                42, n_init=10)
cluster_labels = kmeans.fit_predict(X_pca)
```

Listing 1. K-means

Following clustering, the Silhouette Score was computed to evaluate the cohesion and separation of the resulting clusters. The score obtained was **-0.0016** indicating very weak cluster structure and suggesting that the majority of samples lie near decision boundaries or are poorly assigned.

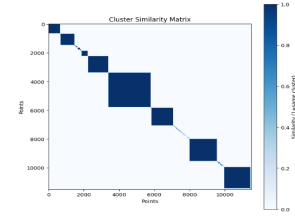


Figure 4. Similarity Matrix

Additionally, a cluster similarity matrix was constructed to visualize pairwise cluster membership. While the matrix exhibits a faint diagonal block pattern consistent with some grouping, the overall lack of strong separation reinforces that K-Means struggled to identify distinct clusters in the high-dimensional, overlapping nature of the EKG signal data. These findings suggest that K-Means may not be well-suited for this type of biomedical time-series data without additional feature engineering or dimensional restructuring.

5.2 Agglomerative Clustering

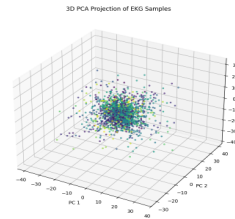


Figure 5. 3D PCA projection of EKG samples

In this section, Agglomerative Clustering was applied to the FFT-transformed EKG signals using Ward's linkage and specifying 115 clusters to match the dataset's subject-activity structure. After fitting the model, the cluster labels were visualized in a 3D scatter plot using PCA with three components. As with previous methods, the plot shows a dense central distribution with significant overlap across clusters, indicating limited separability in the reduced space.

To evaluate the clustering performance, the Silhouette Score was computed and yielded a value of **0.0186**. This score, while slightly better than those from K-Means and GMM, remains very close to zero, suggesting poor cluster compactness and separation. Many samples likely lie near the boundary of their assigned clusters or are indistinguishable from neighboring groups.

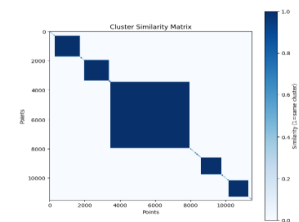


Figure 6. Similarity Matrix

5.3 Gaussian Mixture Models

In this section, the Gaussian Mixture Model (GMM) was applied to the FFT-transformed and standardized EKG dataset using 115 components, corresponding to the number of distinct clusters expected. The model was configured with full covariance and initialized with a fixed random seed to ensure reproducibility.

```
n_components = 115
gmm = GaussianMixture(n_components=
    n_components, covariance_type='full',
    random_state=42)
gmm_labels = gmm.fit_predict(X_fft_filter)
```

Listing 2. GMM

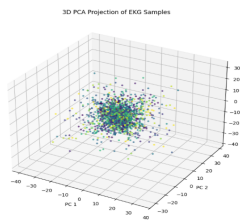


Figure 7. 3D PCA projection of EKG samples

This figure shows the result of applying Principal Component Analysis (PCA) to reduce the dimensionality of the FFT-transformed EKG dataset to three principal components, followed by a 3D scatter plot to visualize the distribution of samples.

To quantitatively assess the clustering quality, the Silhouette Score was computed. The score of **-0.0710** indicates poor clustering. However, it is important to interpret this result with caution in the context of this dataset. The EKG data, derived from continuous biological signals, likely contains high intra-class variability and overlapping frequency characteristics across activities and subjects. Additionally, the structure of the data—where clusters may not be well-separated or convex—violates key assumptions behind the Silhouette Score, which favors distinct, spherical clusters.

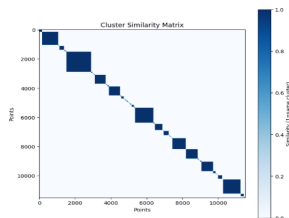


Figure 8. Similarity Matrix

The cluster similarity matrix supports the notion that while there is consistency within certain cluster groupings (evident along the diagonal), there is no strong block-like separation among different clusters. This reinforces the idea that the Silhouette Score may not be a fully reliable metric for assessing clustering performance on high-dimensional, noisy biomedical data such as EKG signals.

6 Results

While the Silhouette Score for Agglomerative Clustering suggests it performs better than the other clustering algorithms tested, it is important to interpret this metric cautiously. In fact, the Gaussian Mixture Model (GMM) achieved the highest Normalized Mutual Information (NMI) score when evaluated against the ground truth labels. At the time of writing this report, the GMM attained an NMI score of 0.3202, placing it first on the leaderboard. This discrepancy highlights a key limitation of the Silhouette Score: it evaluates clustering quality based solely on internal cohesion and separation, assuming well-separated, convex clusters. However, the EKG dataset is high-dimensional and inherently noisy, with significant class overlap and subtle inter-class variations—conditions under which the Silhouette Score becomes unreliable. In such cases, the score may penalize algorithms like GMM that model soft, overlapping clusters, despite their superior performance in capturing latent class structure. Therefore, external validation metrics like NMI, which incorporate ground truth information, provide a more meaningful measure of clustering effectiveness for complex biomedical data.